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Editorial Office, PLOS ONE

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Dear Editors,

I am pleased to submit our manuscript, titled "Is There a Competitive Advantage to Using Multivariate Statistical or Machine Learning Methods Over the Bross Formula in the hdPS Framework for Bias and Variance Estimation?", for consideration as a research article in PLOS ONE.

This study evaluates and compares traditional statistical methods and advanced machine learning approaches within the high-dimensional propensity score (hdPS) framework, focusing on metrics such as bias, mean squared error (MSE), and coverage under diverse epidemiological scenarios. Using a plasmode simulation design and real-world data from the National Health and Nutrition Examination Survey (NHANES), our findings highlight the strengths and limitations of different methods, with implications for selecting appropriate techniques based on the characteristics of the data and research objectives.

Key highlights of our work include:

- A comprehensive comparison of methods, including the kitchen sink model, Bross-based hdPS, LASSO, Elastic Net, Random Forest, and XGBoost, for propensity score estimation and confounding adjustment.
- Demonstrating the advantages and limitations of machine learning approaches in terms of precision and computational efficiency.
- Practical recommendations for selecting hdPS methods tailored to epidemiological scenarios.

This research provides novel insights into the use of statistical and machine learning methods in high-dimensional data settings, offering guidance to researchers in epidemiology, public health, and causal inference.

We believe this manuscript aligns with the scope of PLOS ONE as it contributes to advancing methods for robust and reproducible data analysis in health research. It emphasizes interdisciplinary relevance by bridging statistical innovation and applied public health analysis.

We confirm that:

- The manuscript is original, has not been published elsewhere, and is not under consideration by another journal.
- All authors have read and approved the manuscript and have no conflicts of interest to declare.
- The data supporting our findings are publicly accessible and referenced in the manuscript.

Thank you for considering our submission. We look forward to the opportunity to contribute to PLOS ONE and are happy to address any questions or provide additional information as needed.

Sincerely,

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